

Requirement Document

A restaurant has decided to store information about its daily operations including employees, roles, departments, orders, products, inventory, suppliers, purchases, attendance, and audit logs in a database system. The restaurant has wisely chosen to hire you as a database designer. Prepare an E-R diagram for this Restaurant according to the following description.

- The restaurant has a number of employees. Each employee has **EmployeeID** (unique identifier), **Birth Date**, **Gender**, **FirstName**, **LastName**, **Phone Number**, **Hire Date**, **Salary**, and **Job Title**.
- Each employee must belong to only **one role**. Each role has **RoleID** (unique), **RoleName** (Admin, Cashier, Waiter, Inventory Manager, Manager), and **Description**. Each role may be assigned to several employees.
- Each employee works in only **one department**, and each department may have several employees. Each department has **DeptID** (unique identifier), **DeptName**, and **Location**.
- Each department must have exactly **one manager**. An employee may manage at most one department. The relationship between Department and its Manager includes **Manager Start Date**.
- Each employee has a **supervisor**. One supervisor may supervise several employees, but each employee has only one supervisor.

- The restaurant manages a set of **products**. Each product has **ProductID** (unique identifier), **ProductName**, **Category**, **Price**, **CostPrice**, and **Description**.

- Each product belongs to only **one department**, and each department may manage several products.

- The restaurant records **customer orders**. Each order has **OrderID** (unique identifier), **OrderDate**, **OrderStatus**, **SubTotal**, **Discount**, **Tax**, and **TotalAmount**.

- Each order is created by **one employee** (Cashier or Waiter), and one employee may create several orders.

- Each order may contain several products, and each product may appear in several orders. For each product in an order, the system stores **Quantity** and **Unit Price at the time of sale**.

Note: This requires a junction table **OrderDetails** with

- OrderID, ProductID, Quantity, and UnitPrice.

- The restaurant maintains an **inventory** for products. Each product has exactly **one inventory record**. For each product, the system stores **QuantityInStock**, **MinimumLevel**, and **LastUpdatedDate**.

- The restaurant deals with several **suppliers**. Each supplier has **SupplierID** (unique identifier), **SupplierName**, **Phone**, **Address**, and **Email**.

- Each supplier may supply several products, and each product may be supplied by several suppliers.

Note: This requires a junction table **SupplierProducts** with SupplierID and ProductID.

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- The restaurant records **purchase transactions**. Each purchase has **PurchaseID** (unique identifier), **PurchaseDate**, **TotalAmount**, and **PaymentStatus**. Each purchase must be related to **one supplier**.
 - Each purchase may contain several products, and for each product in a purchase, the system stores **Quantity** and **Unit Cost**.

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- Each employee has **attendance records**. Each attendance record has **AttendanceID** (unique identifier), **WorkDate**, **CheckInTime**, and **CheckOutTime**. Each attendance record must be related to **one employee**, and one employee may have several attendance records.

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- The system stores **audit logs**. Each log has **LogID** (unique identifier), **ActionType**, **ActionDateTime**, and is related to **one employee** who performed the action. One employee may perform several logged actions